Wall time per step: Parse 1M PBMC, full (684,932 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	102	77	71	48	OOM	OOM	40	OOM	OOM	731	OOM	OOM	143
qc	4.0	4.7	3.5	2.3	–	–	7.4	–	–	14	–	–	9.4
normalise	n/a	n/a	n/a	n/a	–	–	0.0	–	–	0.0	–	–	4.6
hvg	1.7	1.7	1.8	1.4	–	–	8.5	–	–	27	–	–	27
pca	4.4	4.5	4.1	1.6	–	–	17	–	–	49	–	–	703
harmony	3.5	3.8	2.5	1.1	–	–	4.5	–	–	254	–	–	218
knn_snn	23	24	12	12	–	–	33	–	–	59	–	–	70
cluster	8.6	8.8	7.1	6.7	–	–	30	–	–	245	–	–	258
cluster_fast	34	37	4.9	3.8	n/a	n/a	n/a	–	–	327	n/a	n/a	n/a
umap	50	50	5.6	5.2	–	–	349	–	–	319	–	–	451
dge	4.5	4.7	4.4	4.1	–	–	–	–	–	TIMEOUT	–	–	1,348
tsne_default	371	373	325	322	–	–	1,941	n/a	n/a	n/a	–	–	1,531
tsne_fft	117	115	20	30	–	–	247	n/a	n/a	n/a	–	–	229
tsne_quick	82	81	42	42	n/a	n/a	n/a	–	–	–	n/a	n/a	n/a
total (no tSNE)	202	180	113	82	failed	failed	490	failed	failed	failed	failed	failed	3,231

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.